

scicomp-research-skills:
Global Agent Skills for Scientific-Computing Research
with two companion tools: `argo-anywhere` + `scrollback`

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argo-anywhere: agents → ANL Argo, any laptop

ANL users only



argo-anywhere

One channel;
one Duo prompt
per session.



[github.com/a-attia/
argo-anywhere](https://github.com/a-attia/argo-anywhere)

`pipx install argo-anywhere`
available on PyPI

```
argo-anywhere connect
[argo-anywhere] Using ANL username: <ANL-username>
[argo-anywhere] Using port: 64742 (source: cached port)
[argo-anywhere] Selected node: <ANL-compute-node>
[argo-anywhere] Opening multiplexed SSH master (Duo prompt expected once)...
[ok] master ready (subsequent SSH calls will reuse this connection)
[argo-anywhere] Copying ansible to <ANL-compute-node>.. argo-anywhere ah
[argo-anywhere] Warning: server bootstrap on <ANL-compute-node>...
[ok] <argo-anywhere> argo-anywhere 2.2.1
[ok] Server is up on <ANL-compute-node>:64742.
[argo-anywhere] Opening tunnel: localhost:64742 => <ANL-compute-node>:64742
[ok] Tunnel is live: argo-anywhere reachable on http://localhost:64742/v1

argo-anywhere -- status summary

ALL GREEN - tunnel up, proxy healthy, 44 models(s)

Connection
Local listener      * pid <pid> bound to 127.0.0.1:64742
Proxy /Auth        * healthy (http://localhost:64742/health)
Tunnel uptime      * 0:11

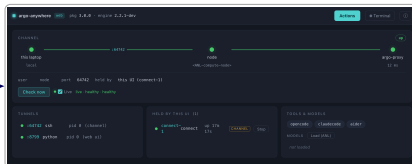
Models
Available          * 44 at /v1/models (e.g. argo-opt-to, argo3d-mini)
OpenCode models    * 23 configured (27 checkable)

Configuration
Port               * 64742
Cached username    * <ANL-username>
Cached node        * <ANL-compute-node>

Next step
Run 'opencode' in another terminal.

[argo-anywhere] Channel is up: point any OpenAI-compatible client at
[argo-anywhere] http://localhost:64742/v1 with Authorization: Bearer
<ANL-username>
```

CLI: *argo-anywhere client* → ALL GREEN



Web UI / native app: *localhost-only*, Duo + in-browser monitor



scrollback: browse / search / export sessions

open to anyone



scrollback

One tool over
every agent's
session store.

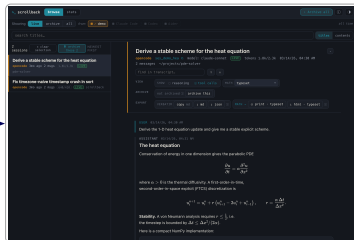


[github.com/a-attia/
scrollback](https://github.com/a-attia/scrollback)

`pipx install scrollback`
`opencode, Claude, Codex, aider`

```
% scrollback list --usage
source  id      updated      msgc  cost  tok in/out  title
opencode ses_demo_hsa 2026-03-14 09:32 2 $0.02 1.8k/2.3k Derive a stable scheme
          for the heat equation
claudecode ses_demo_ref 2026-03-14 08:10 2 $0.01 1.2k/900 Refactor load_config
          with typed validation
opencode ses_demo_bug 2026-03-13 09:32 2 $0.01 640/410 Fix timezone-naive
          timestamp crash in sort
```

CLI: `scrollback list` → one view across all agents



Web UI: rendered Markdown, syntax highlight, typeset LaTeX

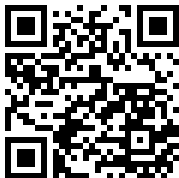


Focus of today's talk: scicomp-research-skills

open to anyone

scicomp-research-skills

Global agent skills + project templates
for scientific-computing research.



github.com/a-attia/scicomp-research-skills

6 skills + 2 project templates – client-agnostic, loaded on demand



Every agent session starts cold

Monday

paper draft

you explain:
LaTeX style, bib
format, notation

Wednesday

numerical library

you explain:
test conventions,
API style, MMS rules

Friday

literature survey

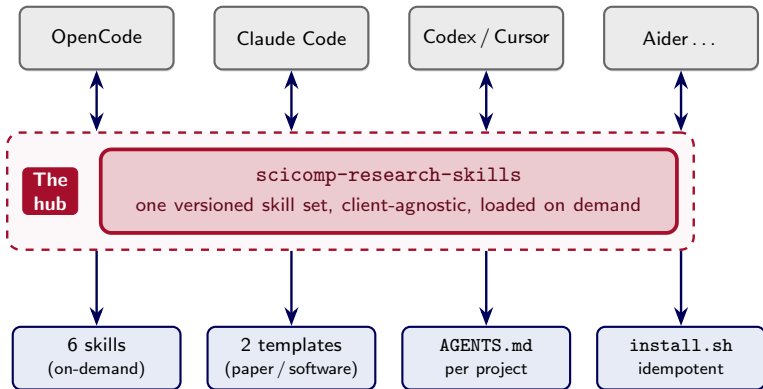
you explain:
survey-note format,
bib verification

**The agent is capable; the conventions keep starting from zero.
Every project. Every session. Every re-open.**

*Three unrelated projects; the agent re-derives
your conventions each time. That's the tax
scicomp-research-skills is designed to remove.*



Teach the agent once. Use the lesson everywhere.



*One markdown-only repo, checked out at
~/.scicomp-research-skills/, discovered automati-
cally by every modern agent client via ~/.agents/skills/.*



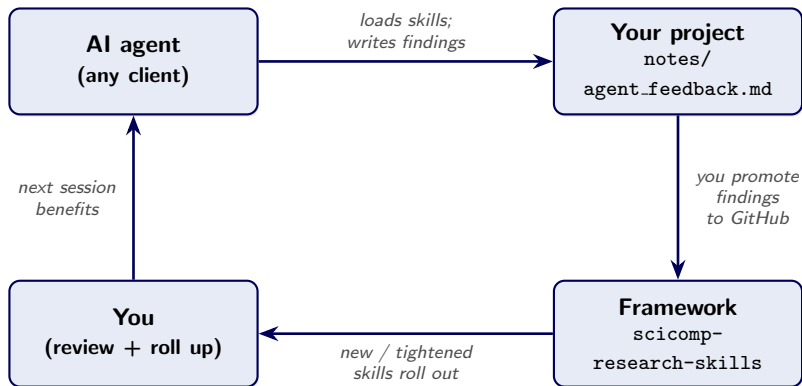
The six skills shipped today

Skill	What it does for the agent
research-paper-writing	Section-by-section drafting; paragraph clarity; claim/evidence alignment; adversarial self-review.
literature-survey	BibTeX + PDF + pdftotext + per-paper survey notes + verification log.
human-facing-doc-authoring	Doc conventions for READMEs, PLANs, notes, rebuttals – anything a human reviews.
agent-resource-discipline	Token/context-window discipline; first- + last-action protocols; persistent cross-session memory.
research-software-engineering	Numerical correctness (MMS, convergence); testing for numerical code; reproducibility; AI-coding rules.
project-onboarding	Inventory-first migration for EXISTING projects; preserves prior agentic files; surfaces conflicts.

Plus two templates: `paper-skeleton/` and `software-skeleton/` – the latter delegates the package layer to community scaffolds (`cookie`, `nlesc`, `wv-cu`, `BestieTemplate.jl`).



Agent, project, framework – a real feedback loop



Every bootstrapped project ships with `notes/agent_feedback.md` + four `.github/ISSUE_TEMPLATE/` skeletons. The loop is the only reason the framework improves at all.



Try it today (I / II): one install

Install once per machine – idempotent; safe to re-run; no root, no PATH edits, no Python dependencies (just git + bash + your existing agent client):

```
git clone git@github.com:a-attia/scicomp-research-skills.git
\
  ~/.scicomp-research-skills \
|| git clone https://github.com/a-attia/scicomp-research-
  skills.git \
  ~/.scicomp-research-skills
~/.scicomp-research-skills/bin/install.sh
```

`install.sh` symlinks the canonical multi-agent path (`~/.agents/skills/`) and each client's own skill directory (`~/.config/opcode/skills/`, `~/.claude/skills/`, ...) into the checkout. Existing paths are warned + skipped, never overwritten. Reverse anytime with `uninstall.sh --confirm` (dry-run by default). See backup slide 2 for the full path map.

Next: once it's installed, using it is a single sentence to your agent →



Try it today (II / II): one sentence to your agent

Once installed, using it is trivial:
tell your agent to adopt *scicomp-research-skills* in your working project.
The framework's project-onboarding skill takes it from there.

Open ANY existing project in your agent client and paste:

```
I want to adopt the scicomp-research-skills framework
(https://github.com/a-attia/scicomp-research-skills) in this project.
If the skills are not installed, install them first -- check the
framework's README for instructions. Then load the project-onboarding
skill, inspect this project, and propose a migration plan before
making any changes.
```

Don't want to copy off my slide? Same prompt lives in the repo's [README](#) "Just want to start?" section. Scan the QR from the pivot slide, open the README, copy the block, paste into your agent. Same result.



Pick one. Install it before you leave the room.



argo-anywhere

companion



scicomp-research-skills

primary today



scrollback

companion

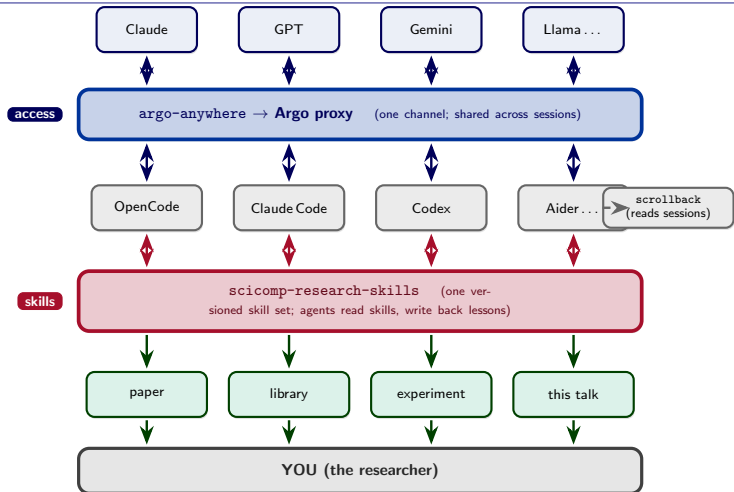
The concrete ask: install **scicomp-research-skills** today, then paste the README's "Just want to start?" prompt into Claude Code on any real project. Send me the first `notes/agent_feedback.md` entry.

Ahmed Attia + Claude (Anthropic AI).

Questions: aattia@anl.gov



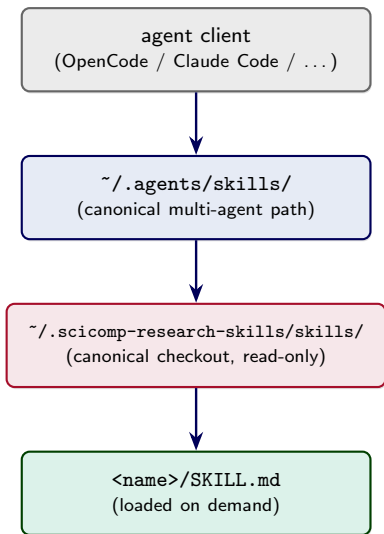
Backup: architecture at a glance



The full stack. All three tools shown together, in the positions they occupy in a daily agent-assisted research workflow at Argonne.



Backup: how a skill is discovered



Wired by `install.sh`:

- Symlinks the canonical multi-agent path
~/.agents/skills/ to the checkout's skills/ directory.
- Also wires per-client paths:
~/config/opencode/skills/,
~/claudio/skills/,
~/cursor/skills/,
~/copilot/skills/,
~/gemini/skills/.
- In-repo filename symlinks for clients that look for a specific name: CLAUDE.md, CONVENTIONS.md, GEMINI.md, .cursorrules, AGENT.md.



Backup: honest STATUS

Well-grounded (trust at face value):

- research-paper-writing – vendored from upstream; has its own user base.
- literature-survey – distilled from a real multi-paper workflow.
- paper-skeleton + software-skeleton – exercised on real projects.

Informed prediction (not yet validated):

- agent-resource-discipline – rules sound; enforcement hooks not yet shipped.
- human-facing-doc-authoring – per-doc-type structures are speculation.
- project-onboarding – ran on 2 projects; 5 sub-scenarios still speculative.

Honest evidence count (2026-05-17):

Signal	Count
Real projects bootstrapped	2
agent_feedback.md lines	~2200
New-skill proposals filed	3
Skill-improvement issues filed	10
External-contributor PRs	0
Roll-up sessions executed	1

2 real-project sessions; both run by the framework author. External-user evidence: still 0.
This deck asks you to be evidence #3.



Backup: why bother? (the three-beat pitch)

- **The problem.** Every new agent session starts cold. The agent is smart; your project's conventions are not in its context. You keep re-explaining them.
- **The idea.** Package the conventions as **on-demand skills** the agent discovers automatically, plus **project templates** that wire a new paper or library into those skills on day one. Every modern coding agent reads the same markdown.
- **The rest of the talk.** What ships today; how it plugs into your daily workflow; one concrete thing to try before the hackathon ends.

*Same three beats structure any short pitch
for `scicomp-research-skills`: name
the tax, name the fix, name the ask.*

